A Giant Hidden Ocean Wave Is Raising Sea Levels on the West Coast

From: NPR World Mined: 2026-10-20 5 min read

A huge wave you cannot see has been raising sea levels along the West Coast of the United States. It is called a **Kelvin wave**, and scientists say more are on the way.

A **Kelvin wave** is not a wall of water or a wave you could surf. "It's a **planetary-scale** wave," said Dillon Amaya, a scientist at North Carolina State University. It moves slowly beneath the ocean's surface and stretches for thousands of miles.

These waves can form along the **equator** during **El Niño** years, when a vast area of the Pacific Ocean becomes much warmer than usual. This year's strong **El Niño** sent warm water east under the sea until it reached South America. There, the wave split. Part of it traveled north along the West Coast, and part went south toward Chile. Columbia University professor Mark Cane compared it to "an enormous bathtub sloshing around."

As the wave moves up the coast, it gradually raises sea levels. Amaya said it does not cause floods by itself, but it makes flooding more likely when a big storm or very high **tides** arrive. In early October, sea levels in parts of Southern California were already about six to eight inches higher than normal. Then Tropical Storm Rachel brought a **storm surge** that flooded some neighborhoods.

Kelvin waves also carry warmer water, which can stress sea life. Tropical fish have shown up in unusual places, and a whale shark was spotted off San Diego.

The wave is named after Lord Kelvin, a British scientist who first described it in the late 1800s. Amaya expects the next **Kelvin wave** to reach San Diego in late October or early November, with several more through the fall and early winter.

Word Treasure

Kelvin wave -- A huge, slow ocean wave that travels below the surface and raises sea levels along coasts.

equator -- The imaginary line around the middle of Earth, halfway between the North and South Poles.

El Niño -- A natural pattern when part of the Pacific Ocean becomes much warmer than usual and changes weather.

planetary-scale -- So large that it affects a big part of the whole planet.

storm surge -- A rise of seawater pushed onto land by a strong storm's winds.

tides -- The regular rise and fall of the sea, mostly caused by the Moon's pull.

Background you need

El Niño is a natural climate pattern in which ocean water in the tropical Pacific becomes much warmer than usual, which can shift weather around the world.

Warm water takes up a little more space than cold water, so when a large mass of it moves along a coast, the sea surface there rises slightly.

Coastal flooding usually happens when several things line up, such as high tides, strong storm winds pushing water ashore and an already higher sea level.

Scientists track these hidden ocean waves with measurements and computer models so coastal communities can prepare for higher flood risk.

Break it down

LEAD: A hidden Kelvin wave is raising sea levels along the U.S. West Coast
KEY EVENT: Warm El Niño water traveled east, split at South America and moved up the coast
+- WHO: Scientists Dillon Amaya and Mark Cane explain the wave
+- WHERE: The West Coast, including Southern California and San Diego
+- WHEN: Now, with the next wave expected in late October or early November
EVIDENCE: Sea levels six to eight inches higher before Tropical Storm Rachel's flooding; tropical fish and a whale shark seen farther north
OPEN QUESTION: How much will the coming waves add to flood risk this fall and winter?

Why it matters

Higher sea levels can make storms more damaging for coastal neighborhoods, and warmer water can stress sea life, so understanding these waves helps communities prepare.

Test what you remember

1. What is a Kelvin wave?

- (A) A tsunami caused by an earthquake
- (B) A surfing wave on beaches
- (C) A wind pattern in the sky
- (D) A slow, huge wave that moves beneath the ocean's surface

2. What ocean pattern helps create these Kelvin waves?

- (A) La Niña
- (B) Monsoon
- (C) The Gulf Stream
- (D) El Niño

3. Where did the warm water travel before splitting?

- (A) To Japan
- (B) To Hawaii
- (C) To South America
- (D) To Antarctica

4. According to Dillon Amaya, how does a Kelvin wave affect flooding?

- (A) It causes floods by itself
- (B) It makes flooding more likely when storms or very high tides arrive
- (C) It has no effect on flooding
- (D) It stops all floods

5. How much higher than normal were sea levels in parts of Southern California in early October?

- (A) About 20 feet
- (B) About six to eight feet
- (C) About one inch
- (D) About six to eight inches

6. Which animal was spotted off San Diego?

- (A) A penguin
- (B) A polar bear
- (C) A walrus
- (D) A whale shark

Answer key (try the quiz first!): 1: D 2: D 3: C 4: B 5: D 6: D

Questions to think about

1. Dillon Amaya, North Carolina State University: Kelvin waves do not cause floods alone but make flooding more likely when storms or high tides arrive.

2. Mark Cane, Columbia University: The wave behaves like an enormous bathtub sloshing around.

MY THOUGHTS (at least 20 words):